

ALGORITHMIC NON-COMMUTATIVE CLASS FIELD THEORY, XX: THE INVARIANT SUBSPACE IN RANK d , AND ONE FLAG COUNT THAT PRODUCES IT

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ABSTRACT. Paper XVIII proved that on an $\tilde{A}_d$ complex of order q the geodesic edge flow satisfies $L_E L_E^\top = q^{d-1}I + q^{d-1}(q-1)T^\top T$ and its dual, so that on any subspace $W \subseteq \ker S \cap \ker T$ invariant under L_E and $L_E^\top$ the restriction is $q^{(d-1)/2}$ times an orthogonal matrix. It left the existence of such a W open, recording only a candidate intertwiner built from the incidence operators $M_i(v, e) = \#\{\text{chambers } C \supseteq e : C_{\tau(e)+i} = v\}$. This note supplies the subspace. The entire content is a single flag count: in $\text{PG}(d, q)$, the number of complete flags F with $F_{i+1} = V$ and F_1 inside a hyperplane H takes exactly *two* values, according as $V \subseteq H$ or not, and their difference $K_i = f(d-i)f(i+1)q^i/[i+1]_q$ depends on neither V nor H . Because the tail of an edge is a hyperplane of the link of its head and the geodesic successors are the points off that hyperplane (Paper XVIII, Lem. 2.2), this dichotomy is exactly an intertwining; and after normalising M_i to the 0/1 indicator N_i the constant collapses to a bare power, $K_i/c_i = q^i$, leaving

$$L_E N_i^\top = q^i (N_0^\top A_{i+1} - N_{i+1}^\top), \quad N_0 = T, \quad N_d = S,$$

with the mirror identity for $L_E^\top$ from the self-duality of $\text{PG}(d, q)$. Hence $\Psi_d = [N_0^\top \cdots N_d^\top]$ satisfies $L_E \Psi_d = \Psi_d C$ and $L_E^\top \Psi_d = \Psi_d C'$, and $W_d = (\text{im } \Psi_d)^\perp$ is a two-sided invariant subspace of $\ker S \cap \ker T$. Paper XVIII's Corollary 2.7 therefore holds with no hypothesis: every eigenvalue of $L_E|_{W_d}$ has modulus exactly $q^{(d-1)/2}$. The companion matrix is explicit in every rank and its characteristic polynomial is $\sum_j (-1)^j q^{j(j-1)/2} A_j x^{d+1-j}$, the local Hecke polynomial of PGL_{d+1} — the cubic pencil of Kang and Li at $d=2$, and at $d=3$ the quartic $x^4 - A_1 x^3 + q A_2 x^2 - q^3 A_3 x + q^6$, obtained with no global $\tilde{A}_3$ complex. The relations $N_i^\top 1_{t+i} = \binom{d}{i}_q N_0^\top 1_t$ give $\text{corank } \Psi_d \geq d(d+1)$, with equality at $d=2$. The upshot is a decomposition of the spectrum of L_E into a Hecke part on $\text{im } \Psi_d$, where the Ramanujan property is exactly the question, and a complement on W_d that is on the critical circle unconditionally. Eight checks, verified on $\text{PG}(d, q)$ for fourteen triples (d, q, i) , symbolically in q for $d \leq 7$, and on the three thick $\tilde{A}_2$ complexes of Paper X.

1. THE GAP, AND WHAT CLOSES IT

Let X be a thick $\tilde{A}_d$ complex of order q : a finite quotient of the Bruhat–Tits building of PGL_{d+1} over a local field with residue field of size q , with vertices typed by $\mathbb{Z}/(d+1)$. Write E for the set of directed edges $x \rightarrow y$ with $\tau(y) = \tau(x) + 1$, S and T for the tail and head incidence matrices, and L_E for the geodesic edge flow,

$$(L_E)_{e,f} = 1 \iff e = (x, y), f = (y, z) \text{ and } \{x, y, z\} \text{ is not a simplex of } X.$$

Paper XVIII proved

$$(1) \quad L_E L_E^\top = q^{d-1}I + q^{d-1}(q-1)T^\top T, \quad L_E^\top L_E = q^{d-1}I + q^{d-1}(q-1)S^\top S,$$

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for every $d \geq 2$, and deduced that on any subspace $W \subseteq \ker S \cap \ker T$ stable under both L_E and $L_E^\top$ the restriction $R = L_E|_W$ satisfies $RR^\top = R^\top R = q^{d-1}I_W$, so that all its eigenvalues have modulus $q^{(d-1)/2}$ — with no Ramanujan hypothesis. What it could not do was produce a single such W . Its Remark 4.1 recorded a candidate and observed that the identities it would need are “entrywise local”, so that they could in principle be settled inside $\text{PG}(d, q)$ without a global complex. That is what is carried out here, and the local computation turns out to be one line.

Definition 1. For $0 \leq i \leq d$ and an edge $e = (x, y) \in E$ put

$$N_i(v, e) = \begin{cases} 1, & \{x, y, v\} \text{ lies in a chamber and } \tau(v) = \tau(y) + i, \\ 0, & \text{otherwise,} \end{cases}$$

and $\Psi_d = [N_0^\top \ N_1^\top \ \cdots \ N_d^\top] \in \mathbb{Z}^{E \times (d+1)n}$. Write c_i for the number of chambers containing such a triple, so that the unnormalised operator of Paper XVIII, Rem. 4.1 is $M_i = c_i N_i$.

Two blocks are old: $N_0 = T$ and $N_d = S$, because a vertex of relative type 0 in a chamber through $e = (x, y)$ is y and one of relative type d is x . So $\text{im } \Psi_d \supseteq \text{im } S^\top + \text{im } T^\top$ and therefore

$$(2) \quad W_d := (\text{im } \Psi_d)^\perp \subseteq (\text{im } S^\top)^\perp \cap (\text{im } T^\top)^\perp = \ker S \cap \ker T.$$

So the only thing to prove is that $\text{im } \Psi_d$ absorbs L_E and $L_E^\top$.

2. ONE FLAG COUNT

Write $[j]_q = (q^j - 1)/(q - 1)$ for the number of points of a j -dimensional $\mathbb{F}_q$ space and $f(m) = \prod_{j=1}^m [j]_q$ for the number of complete flags of an m -dimensional one. A *complete flag* of $\text{PG}(d, q)$ is a chain $F_1 \subset F_2 \subset \cdots \subset F_d$ of subspaces of an ambient $(d+1)$ -dimensional space, $\dim F_j = j$.

Theorem 2. Fix $0 \leq i \leq d-1$, a hyperplane H and a subspace V with $\dim V = i+1$. Then

$$N(V, H) := \#\{\text{complete flags } F : F_{i+1} = V \text{ and } F_1 \subseteq H\} = \begin{cases} f(i+1)f(d-i), & V \subseteq H, \\ f(i+1)f(d-i) - K_i, & V \not\subseteq H, \end{cases}$$

where

$$K_i = \frac{f(d-i)f(i+1)q^i}{[i+1]_q}$$

is a positive integer depending only on d, i and q . In particular N takes exactly two values, and the jump is independent of both V and H .

Proof. A complete flag with $F_{i+1} = V$ is a complete flag of V together with a complete flag of the quotient by V , and the two choices are independent; the quotient has dimension $(d+1) - (i+1) = d-i$, so the number of flags above V is $f(d-i)$ regardless of H .

Below V the constraint is $F_1 \subseteq H \cap V$. If $V \subseteq H$ this is vacuous and the count is $f(i+1)$. Otherwise $H \cap V$ is a hyperplane of V , of dimension i . Every point of V lies in the same number $f(i+1)/[i+1]_q$ of complete flags of V , by transitivity of $\text{GL}(V)$ on points; hence the count is $[i]_q f(i+1)/[i+1]_q$, and

$$f(i+1) - \frac{[i]_q f(i+1)}{[i+1]_q} = f(i+1) \frac{[i+1]_q - [i]_q}{[i+1]_q} = \frac{f(i+1)q^i}{[i+1]_q}.$$

Multiplying by $f(d-i)$ gives K_i . It is an integer because it counts flags. $\square$

d	q	i	$N(V \subseteq H)$	$N(V \not\subseteq H)$	K_i
2	2	0		3	0
2	2	1		3	1
2	3	0		4	0
2	3	1		4	1
3	2	1		9	3
3	2	2		21	9
3	3	1		16	4
3	3	2		52	16
4	2	1		63	21
4	2	2		63	27
4	2	3		315	147
					168

TABLE 1. The two values of $N(V, H)$ and their difference, over all pairs (V, H) in $\text{PG}(d, q)$. The rows with $i = 0$ are omitted for $d \geq 3$ (there V is a point, $N = 0$ when $V \not\subseteq H$, and $K_0 = f(d)$). Checks (F) and (K).

Theorem 2 is verified exhaustively — over *every* pair (V, H) — for (d, q) equal to $(2, 2)$, $(2, 3)$, $(3, 2)$, $(3, 3)$ and $(4, 2)$ and every i , fourteen triples in all; checks (F) and (K). Table 1 lists the values.

3. THE INTERTWINING

Lemma 3 (Paper XVIII, Lem. 2.2). *Let y be a vertex of X and identify its link with the flag complex of $\text{PG}(d, q)$ so that a neighbour of relative type $+j$ corresponds to a subspace of dimension j . Then the tail of an edge with head y is a hyperplane, the geodesic successors are exactly the points not on it, and the chambers of X through y are the complete flags.*

Theorem 4. *Let $A_j(y, v) = 1$ when v is a neighbour of y of relative type $+j$ and 0 otherwise, and put $N_{d+1} := 0$, $A_{d+1} := I$. Then for every $\tilde{A}_d$ complex of order q and every $0 \leq i \leq d$,*

$$(3) \quad L_E N_i^\top = q^i (N_0^\top A_{i+1} - N_{i+1}^\top),$$

and dually, with $N_{-1} := 0$ and $A_0 := I$, $L_E^\top N_i^\top = q^{d-i} (N_d^\top A_i - N_{i-1}^\top)$. Consequently $L_E \Psi_d = \Psi_d C$ and $L_E^\top \Psi_d = \Psi_d C'$ for $(d+1) \times (d+1)$ block matrices with entries in the algebra generated by $A_1, \dots, A_d$.

The collapse of the constant from K_i to the bare power q^i is what makes the rank- d statement usable, and it is forced: a chamber through $\{x, y, v\}$ is a flag through the chain $V \subset H$ of dimensions $i \subset d$, so $c_i = f(i)f(d-i)$, and with $f(i+1) = [i+1]_q f(i)$,

$$(4) \quad \frac{K_i}{c_i} = \frac{f(d-i)f(i+1)q^i}{[i+1]_q} \cdot \frac{1}{f(i)f(d-i)} = q^i.$$

Proof. Fix $e = (x, y)$ and a vertex v of type $\tau(y) + 1 + i$. Then

$$c_i (L_E N_i^\top)(e, v) = \sum_{z: (y, z) \text{ geodesic after } e} M_i(v, (y, z)) = \#\{(z, C) : z \notin x, y, z \in C, C_{\tau(y)+1+i} = v\},$$

the condition $z \notin x$ being Lemma 3. Work in the link of y . There x is a hyperplane H , each z is a point p , v is a subspace V of dimension $i+1$, and a chamber C through y and z with

$C_{\tau(y)+1+i} = v$ is a complete flag F with $F_1 = p$ and $F_{i+1} = V$. Since z is recovered from C as F_1 , the pair (z, C) is just the flag, and

$$c_i(L_E N_i^\top)(e, v) = \#\{F : F_{i+1} = V\} - \#\{F : F_{i+1} = V, F_1 \subseteq H\} = f(i+1)f(d-i) - N(V, H).$$

By Theorem 2 this is 0 when $V \subseteq H$ and K_i otherwise. Now $V \subseteq H$ says precisely that v and x lie in a common chamber with y , i.e. that $\{x, y, v\}$ is a simplex, which is exactly what N_{i+1} records; and the value K_i is attained only when v is a neighbour of y of relative type $i+1$ at all, which is $A_{i+1}(y, v)$. Assembling and dividing by c_i , which is q^i by (4),

$$(L_E N_i^\top)(e, v) = q^i (A_{i+1}(y, v) - N_{i+1}(v, e)),$$

and $A_{i+1}(y, v) = (N_0^\top A_{i+1})(e, v)$ since $N_0 = T$ and y is the head of e . At $i = d$ the first term is $q^d N_0^\top$ and the second vanishes, recovering $L_E S^\top = q^d T^\top$, which is Paper XVIII, Prop. 2.4 transposed.

For the dual, $(L_E^\top N_i^\top)(f, v)$ with $f = (y, z)$ sums $N_i(v, (x, y))$ over the x with $\{x, y, z\}$ not a simplex. In the link of y this counts complete flags F with $F_i = V$ and $p \not\subseteq F_d$, where now V has dimension i and p is the point z . The duality of $\text{PG}(d, q)$, which exchanges subspaces of dimension j with those of dimension $d+1-j$ and reverses inclusion, carries this count to the count of Theorem 2 with i replaced by $d-i$. The same argument then gives the mirror identity, with $N_d = S$ in place of $N_0 = T$ and N_{i-1} in place of N_{i+1} .

Finally, each right-hand side of (3) is a $\mathbb{Z}$ -linear combination of the blocks $N_0^\top$ and $N_{i+1}^\top$, the first multiplied on the right by an operator on the vertex space. That is exactly the statement $L_E \Psi_d = \Psi_d C$ with C of the asserted shape. $\square$

Corollary 5. $W_d = (\text{im } \Psi_d)^\perp$ is contained in $\ker S \cap \ker T$ and is invariant under L_E and $L_E^\top$. Hence, by Paper XVIII, Cor. 2.7, $R = L_E|_{W_d}$ satisfies $RR^\top = R^\top R = q^{d-1}I$, every eigenvalue of R has modulus $q^{(d-1)/2}$, and every root of $\det(I - uR)$ lies on $|u| = q^{-(d-1)/2}$. No hypothesis on X is used — in particular, no Ramanujan hypothesis.

Proof. The inclusion is (2). If $L_E \Psi_d = \Psi_d C$ then $L_E(\text{im } \Psi_d) \subseteq \text{im } \Psi_d$, so $L_E^\top$ preserves $(\text{im } \Psi_d)^\perp$; and $L_E^\top \Psi_d = \Psi_d C'$ gives the other half. Now apply (1), whose correction terms $T^\top T$ and $S^\top S$ vanish on W_d . $\square$

4. THE COMPANION MATRIX, AND THE HECKE POLYNOMIAL IN RANK d

Reading (3) columnwise gives C outright: the block in row 0, column i is $q^i A_{i+1}$ and the block in row $i+1$, column i is $-q^i I$, all others zero.

Theorem 6. $L_E \Psi_d = \Psi_d C$ with

$$C = \begin{pmatrix} A_1 & qA_2 & q^2A_3 & \cdots & q^{d-1}A_d & q^dI \\ -I & 0 & 0 & & & 0 \\ 0 & -qI & 0 & & & 0 \\ & & \ddots & & & \vdots \\ 0 & 0 & & -q^{d-1}I & 0 & 0 \end{pmatrix},$$

and, the blocks $A_1, \dots, A_d$ being commuting operators on the vertex space,

$$\det(xI - C) = \sum_{j=0}^{d+1} (-1)^j q^{j(j-1)/2} A_j x^{d+1-j}, \quad A_0 = A_{d+1} = I,$$

the local Hecke polynomial of PGL_{d+1} . Equivalently, on $\text{im } \Psi_d$,

$$L_E^{d+1} - L_E^d A_1 + qL_E^{d-1} A_2 - q^3 L_E^{d-2} A_3 + \cdots + (-1)^{d+1} q^{d(d+1)/2} = 0.$$

Proof. The shape of C is (3). For the determinant, eliminate: (3) gives $N_{i+1}^\top = N_0^\top A_{i+1} - q^{-i} L_E N_i^\top$, so by induction $N_j^\top = \sum_{t=0}^j (-1)^t q^{-(tj-t(t+1)/2)} L_E^t N_0^\top A_{j-t}$. Applying L_E to the case $j = d$ and using $L_E N_d^\top = q^d N_0^\top$ gives a relation of degree $d+1$ in L_E annihilating $N_0^\top$; clearing the negative powers of q turns it into the displayed polynomial, whose coefficient of $L_E^{d+1-j} A_j$ is $(-1)^j q^{j(j-1)/2}$. $\square$

At $d = 2$ this is $x^3 - A_1 x^2 + q A_2 x - q^3$, the cubic pencil of Kang and Li [3]. At $d = 3$, which no published work reaches, it is

$$C^{(4)} = \begin{pmatrix} A_1 & qA_2 & q^2 A_3 & q^3 I \\ -I & 0 & 0 & 0 \\ 0 & -qI & 0 & 0 \\ 0 & 0 & -q^2 I & 0 \end{pmatrix}, \quad \det(xI - C^{(4)}) = x^4 - A_1 x^3 + q A_2 x^2 - q^3 A_3 x + q^6,$$

the quartic local Hecke polynomial of PGL_4 — obtained here with no global $\tilde{A}_3$ complex, purely from the flag count. The determinant identity is verified symbolically for $d = 2, 3, 4, 5$, and (4) for $d \leq 6$; checks (H) and (A).

Theorem 7. *For every $t \in \mathbb{Z}/(d+1)$ and every i , let $1_t \in \mathbb{Z}^n$ be the indicator of the vertices of type t , and let $b_i = \binom{d}{i}_q = f(d)/(f(i)f(d-i))$ be the number of vertices of relative type i forming a simplex with a given edge. Then*

$$N_i^\top 1_{t+i} = b_i N_0^\top 1_t,$$

so the vectors $w = (w_0, \dots, w_d)$ with $w_i = \sum_s a_{i,s} 1_s$ lie in $\ker \Psi_d$ exactly when $\sum_{i=0}^d b_i a_{i,t+i} = 0$ for each t . That is $(d+1)^2$ unknowns subject to $d+1$ independent equations, so

$$\mathrm{corank} \Psi_d \geq d(d+1), \quad \dim W_d \leq |E| - (d+1)n + \mathrm{corank} \Psi_d.$$

Proof. $(N_i^\top 1_s)(e) = \sum_{\tau(v)=s} N_i(v, e)$ counts the vertices of relative type i lying in a chamber with e , when $s = \tau(\mathrm{head} e) + i$, and is 0 otherwise. In the link of the head those vertices are the i -dimensional subspaces of the hyperplane x , so there are $b_i = \binom{d}{i}_q$ of them, independently of e ; hence $N_i^\top 1_s = b_i N_0^\top 1_{s-i}$. The $d+1$ vectors $N_0^\top 1_t$ are the indicators of the type classes of E and so are independent, which makes the $d+1$ equations independent. $\square$

On all three complexes of Section 5 the corank is exactly $d(d+1) = 6$ and these relations span the kernel, so $\dim W_d = |E| - (d+1)n + d(d+1)$ there; check (N). Whether the inequality is an equality for $d \geq 3$ we do not know. We note that $d(d+1)$, $2(d+1)$ and $(d+1)!$ all equal 6 at $d = 2$ and diverge at $d = 3$, where they are 12, 8 and 24; the value derived above is $d(d+1)$.

5. RANK TWO, AND THE HECKE PENCIL

At $d = 2$ we have $N_0 = T$, $N_2 = S$ and $N_1 = M$, the chamber indicator; the three cases of (3) read

$$(5) \quad L_E T^\top = T^\top A_1 - M^\top, \quad L_E M^\top = q(T^\top A_2 - S^\top), \quad L_E S^\top = q^2 T^\top,$$

with the mirror obtained by exchanging $S \leftrightarrow T$ and $A_1 \leftrightarrow A_2$. Only the first and third were available before: the first is the definition of the geodesic condition rearranged, the third is Paper XVIII, and the middle one — the one that closes the cycle — is new. With $\Psi_2 = [T^\top \ M^\top \ S^\top]$,

$$C = \begin{pmatrix} A_1 & qA_2 & q^2 I \\ -I & 0 & 0 \\ 0 & -qI & 0 \end{pmatrix}, \quad \det(xI - C) = x^3 - A_1 x^2 + q A_2 x - q^3.$$

X	n	$ E $	q	$\text{rank } \Psi_2$	$\dim W_2$	eigenvalues of $L_E _{W_2} $
Y_{21}	21	147	2	57	90	$1.414214 = \sqrt{2}$
Y_{24}	24	168	2	66	102	$1.414214 = \sqrt{2}$
Y_{42}	42	294	2	120	174	$1.414214 = \sqrt{2}$

TABLE 2. The invariant subspace on the three thick $\tilde{A}_2$ complexes of Paper X. In every case Ψ_2 has corank 6 in $\mathbb{Z}^{3n}$ — the two relations per type saying that the columns of S , T and M sum to the indicator of a type class — and *every* eigenvalue of $L_E|_{W_2}$ has modulus $\sqrt{q}$ to machine precision. Checks (P) and (U).

That cubic is the local Hecke polynomial of PGL_3 , and the determinant identity built on it is Kang and Li's [3]. The three identities of (5), their mirrors, $L_E\Psi_2 = \Psi_2C$ exactly, and the pencil relation $C^3 = A_1C^2 - qA_2C + q^3$ are verified on the thick $\tilde{A}_2$ complexes Y_{21} , Y_{24} and Y_{42} of Paper X; checks (I) and (P). Table 2 records the subspace they produce.

Remark 8 (What the decomposition says). Corollary 5 splits the spectrum of L_E into a part carried by $\text{im } \Psi_d$ and a part carried by W_d . On $\text{im } \Psi_d$ the flow is governed by C , whose entries are Hecke operators, so that part of the spectrum is a function of the Satake parameters and the Ramanujan property is exactly the question of where it sits. On W_d the flow is $q^{(d-1)/2}$ times an orthogonal matrix, unconditionally. At $d = 2$ this explains the shape of Kang and Li's criterion [3, Thm. 1.0.2]: they prove that X is Ramanujan if and only if the non-trivial zeros of $\det(I - uL_E)$ have $|u| \in \{q^{-1}, q^{-1/2}\}$. The $q^{-1/2}$ family is precisely the W_d part, and Corollary 5 says it is there for every complex, Ramanujan or not; the Ramanujan content is entirely in the q^{-1} family, which is the Hecke part.

Remark 9 (Solved, open, impossible). *Solved*: the open problem (a) of Paper XVIII, Rem. 4.2 — the invariant subspace, and with it Corollary 2.7 of that paper unconditionally, for every $d \geq 2$. *Open*: (i) whether the bound $\text{corank } \Psi_d \geq d(d+1)$ of Theorem 7 is an equality for $d \geq 3$, hence $\dim W_d$ exactly; at $d = 2$ it is an equality on all three complexes tested, and we have not proved it in general. (ii) Whether $\text{im } \Psi_d$ is the *full* Hecke-isotypic part, i.e. whether W_d is the largest two-sided invariant subspace inside $\ker S \cap \ker T$. (iii) A rank- d analogue of the determinant identity itself; see the provenance note. *Impossible by the same argument*: nothing here bounds the spectrum on $\text{im } \Psi_d$. The Hecke part is where Ramanujan lives, and Corollary 5 is silent about it by construction — that is the price of being unconditional.

6. VERIFICATION

The script `ad_invariant_subspace.py` runs eight checks in a few seconds; its output is archived as `ad_invariant_subspace_output.txt`.

Check (F) enumerates every complete flag of $\text{PG}(d, q)$ and every pair (V, H) — no sampling — for the five pairs (d, q) with at most 20 000 flags. Check (U) is the only numerical item; it is a consequence of the exact checks, and is reported because the eigenvalues are what the statement is about.

7. SCOPE AND PROVENANCE

Scope. Theorem 2 and Theorem 4 are proved for every $d \geq 2$ and every $\tilde{A}_d$ complex of order q ; so is Corollary 5. The global identities are *verified* only at $d = 2$, because we have no $\tilde{A}_d$ complex for $d \geq 3$ to verify them on; the proof of Theorem 4, however, uses nothing beyond Lemma 3 and the flag count, and both hold in every rank. The corank of Ψ_d is not determined.

key	claim	cases	mode
(F)	$N(V, H)$ takes exactly two values, jump K_i	14 triples (d, q, i)	exact
(K)	the closed forms f, φ, K_i match the enumeration	14 triples	exact
(A)	$K_i/c_i = q^i$, symbolically in q	$2 \leq d \leq 7$, all i	exact
(H)	$\det(xI - C)$ is the PGL_{d+1} Hecke polynomial	$d = 2, 3, 4, 5$	exact
(I)	the six rank-2 identities of (5) and their mirrors	3 complexes	exact
(P)	$\mathrm{im} \Psi_2$ absorbs L_E and $L_E^\top$; C obeys the pencil	3 complexes	exact
(U)	every eigenvalue of $L_E _{W_2}$ has modulus $q^{1/2}$	3 complexes	numeric
(N)	$\mathrm{corank} \Psi_2 = 6 = d(d+1)$, and the type relations span	3 complexes	exact

TABLE 3. The battery. “Exact” is exact integer arithmetic; (U) is a floating-point eigenvalue computation, agreeing with $\sqrt{q}$ to 10^{-12} .

Provenance. Identity (1) and Lemma 3 are Paper XVIII. The $d = 2$ determinant identity $Z(X, u)^{-1} = \det(I - A_1 u + q A_2 u^2 - q^3 u^3) \cdot \det(I + L_B u) \cdot (1 - u^3)^{-\chi}$ and the four-way Ramanujan criterion quoted in Remark 8 are Kang and Li [3]; a representation-theoretic reproof is Kang, Li and Wang [4], and an earlier approximation is Deitmar and Hoffman [1]. The rank-two case for Sp_4 is Fang, Li and Wang [2]. For general rank the state of the art is an alternating-product Ihara identity, uniform in n , of Kang and Yu [6]; it carries no Riemann-hypothesis statement and no single-operator determinant formula, and Kang and McCallum [5] had recorded the higher-rank case as a conjecture. A targeted search found no published statement of an *operator-level* intertwining $L_E \Psi = \Psi C$ — at any rank, including $d = 2$ — and none of an unconditional modulus statement for a complementary part of $\mathrm{spec}(L_E)$. Those two, and the flag count that produces them, are what is new here. The spectral definition of a Ramanujan complex of type $\tilde{A}_d$ is Lubotzky, Samuels and Vishne [7].

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